

SYNOPSIS

At about 1125 on 17 September 2019, two fire and rescue service boats - a 4.7m Zodiac Milpro inflatable boat and a 6.4m Delta rigid inflatable boat - were in collision while undertaking boat training and familiarisation on the River Cleddau near Milford Haven, Wales. The collision resulted in one of the firefighters who was taking part in the training on board the Zodiac, being struck by the Delta RIB and sustaining fatal injuries. The collision occurred because both boats were operating at speed and carrying out un-coordinated manoeuvres in the same stretch of the river. The Delta RIB's helmsman attempted to pass between the Zodiac and the shoreline as his boat exited a large turn made at speed. The Zodiac's helmsman, initially unaware of the Delta RIB approaching, turned his boat sharply to port as part of his training manoeuvres, which resulted in the boats heading towards each other. The subsequent action taken by both boats to avoid a collision was unsuccessful.

The MAIB investigation found that the accident could have been avoided had the training activities been properly planned. Furthermore:

- No-one was in overall charge of the training and familiarisation activities, so they were not properly managed, briefed or communicated between the crews of both boats.
- The operation of the boats did not adhere to the requirements of the local standard operating procedures or risk assessments.
- The standard operating procedures for all fire and rescue service boats in the Mid and West Wales Fire and Rescue Service were insufficient in content and contained incorrect information.
- The generic risk assessment for boat operations was inadequate.
- The Fire and Rescue Services in the United Kingdom did not operate boats to a common standard or code of practice when not employed on flood rescue duties.

Mid and West Wales Fire and Rescue Service has undertaken its own internal investigation; has implemented a peer review by two other fire and rescue service authorities for various activities, including boat operations; and has reviewed and updated its standard operating procedures and risk assessments for fire service boats. Mid and West Wales Fire and Rescue Service has been recommended to review the crewing and staff qualification requirements for its boats, implement measures to maintain crew competency as boat helmsman, and improve its auditing process to include its boat operations. The National Fire Chief's Council has been recommended to adopt a nationwide standard of boat operations for all craft capable of being operated at sea by any fire and rescue service across the United Kingdom.

1.3 ENVIRONMENTAL CONDITIONS

The wind was variable in direction and had a strength of force 1 to 2. It was a sunny day with very few clouds. The temperature was about 19°C and the visibility was excellent.

High water at Neyland was at 0828 with a height of 6.72m, and low water was at 1430 with a height of 1.10m.

1.4 NARRATIVE

At about 0800 on 17 September 2019, the 10 firefighters assembled at Milford Haven fire station and were split into two teams of five. One team was chosen to go out in the boats during the morning while the other manned the fire station. The plan was to swap the teams at lunchtime. At about 0830, following a short discussion, the morning boat team was split into two groups. One of the firefighters was also a WRBO instructor (hereafter referred to as the trainer) and he and the two WRBO trainees were assigned to *N207*. The two remaining firefighters, each of whom held the rank of crew manager, were assigned to *Rescue 1*.

N207 was made ready at the fire station, then towed down to a launch site at Gelliswick (**Figure 3**). When they arrived, the three firefighters donned their dry-suits, boots and Palm Rescue 800 buoyancy aids. The boat was equipped with Nitro XT water safety helmets, but they were not worn by the crew. At 0938, the firefighters launched and boarded *N207*. One of the firefighters secured the engine kill-cord to his wrist and took the helm; the trainer positioned himself in the middle of the boat and the other firefighter sat in the bow. In the meantime, the towing vehicle and trailer returned to the fire station to collect *Rescue 1*.

At about 0945, *N207* left Gelliswick and was steered eastwards through the Milford Haven waterway and toward the Cleddau bridge. While en route the trainer pointed out different types of buoyage to the firefighters, discussed tides, and talked about his plans for the morning. At the bridge, the trainer asked the firefighters to practise some swift water boat manoeuvres. However, the flow of the ebbing tide at the bridge was insufficient to practise some intended rope and engine work, so they moved the boat a short distance upstream to an old fish farm structure on the south shore where the tide was ebbing with more strength. The firefighters then swapped places and the second firefighter took over the helm.

At 1050, *Rescue 1* was launched with the two crew managers on board. Each was wearing a dry-suit, boots and Crewsaver Offshore 290N inflatable lifejacket. *Rescue 1* was equipped with Geko open face marine safety helmets, but they were not worn by either crew member. The helmsman positioned himself on the port side of the boat behind the steering wheel, and secured the engine kill-cord to his lifejacket. The other crew manager, who was seated on the starboard side next to the helmsman, assumed the role of crewman and used a hand-held VHF radio to inform the local coastguard that *Rescue 1* was on the water. He also explained that they would be engaged in training activities in the eastern part of Milford Haven waterway and in the River Cleddau.

Rescue 1's helmsman steered the RIB eastwards, under the Cleddau bridge, and met up with *N207* at the disused fish farm structure. During a shouted discussion between the two boat crews it was agreed that each would make their way independently upstream, practising operating the boats at high speed, and rendezvous again at Mill Bay on the south shore (**Figure 3**).

As there were no speed restrictions in that part of the river, both helmsmen proceeded ahead at, or close to, their boat's full speed. *Rescue 1*, which was travelling at a speed of about 28kts, arrived at Mill Bay ahead of *N207*, which was being driven at its full speed of 16kts.

By about 1115, both boats had arrived at Mill Bay and were rafted together, drifting about 20m from the south shore and pointing upstream. As they drifted, the crews discussed the local topography, the nature of the riverbed and local depths of water. During the conversation they agreed to proceed further upstream to the area of Lawrenny and rendezvous again there.

Note: The accounts of the witnesses with regard to the events after the boats left Mill Bay are not wholly consistent, and this section of the narrative is based upon the investigator's understanding of the most likely sequence of events.

Shortly after 1120, the two boats parted, with *Rescue 1* proceeding ahead and accelerating to about half full speed. Meanwhile, *N207* proceeded upstream at variable speeds, with its helmsman sitting on the starboard side of the boat practising manoeuvres that involved large course alterations. *N207*'s trainer was sitting in the middle of the boat with his back against the port side tube, facing to starboard. The other firefighter, Josh Gardener, was sitting in the bow on the port side (**Figure 4**).

Once ahead and clear of *N207*, *Rescue 1*'s helmsman commenced a large full-circle turn of the boat to port (**Figure 5a**). As the boat neared the end of its final turn to port travelling at about 15kts, the helmsman saw *N207* ahead of him in the distance, slightly to port, and noted that it now appeared to be headed downstream towards Mill Bay (**Figure 5b**). He decided to pass between *N207* and the shoreline. As *Rescue 1* came out of its turn, the helmsman's attention was briefly taken away from *N207* and was split between operating the steering in front of him, checking the engine gauges, and controlling the throttles.

For illustrative purposes only: not to scale

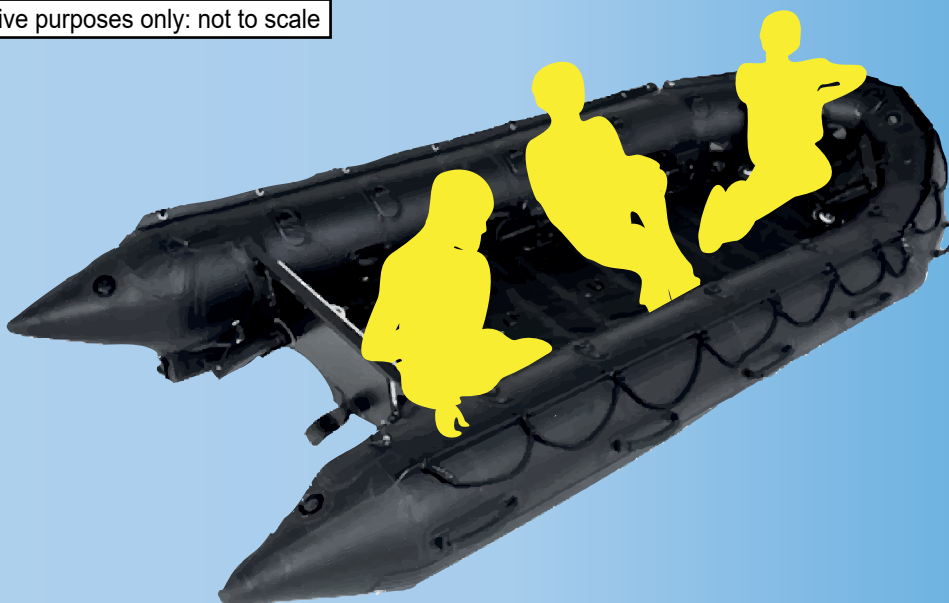


Figure 4: Location of crew in *N207*

Rescue 1's bow struck *N207* in the middle of the boat at an angle of about 45° and rode over its buoyancy tube toward its bow (**Figure 6a**). *Rescue 1*'s helmsman pulled his kill-cord, which stopped the boat's engines. The force of the impact threw *N207*'s helmsman forward, which pulled out his kill-cord and stopped the boat's engine. The firefighter in the bow of *N207* was knocked or thrown into the water. He was struck on the head during the collision and as the boats separated was seen floating motionless on the water.

1.5 EMERGENCY RESPONSE

N207's trainer immediately jumped into the water to attempt to rescue the casualty, but it quickly became apparent to him, from the severity of the injuries, that the firefighter was dead.

Rescue 1 was manoeuvred alongside the casualty in the water, and his body was recovered onto the boat using the onboard casualty recovery equipment. The trainer climbed back into *N207* and brought it alongside *Rescue 1*. He then transferred into the bigger boat with *N207*'s remaining crewman to assist.

At 1127, the crew manager who had been acting as crew on *Rescue 1* made a telephone call to the fire station watch manager and told him what had happened. At 1129, he called the emergency services and reported the accident.

While the phone calls were being made, *N207* was placed under the casualty recovery equipment and secured alongside *Rescue 1*. The deceased's body was then lowered into the smaller boat, placed in a body bag, and covered over. *Rescue 1* then motored slowly downstream toward Neyland yacht club.

At 1145, while on passage downstream, the crewman on board *Rescue 1* received a call from the local ambulance service to verify that attempts at resuscitation could not be carried out by the firefighters, and to inform them that they would be met on arrival.

At 1155 *Rescue 1* arrived at Neyland yacht club pontoon, where they were met by members of the police, ambulance service and fire and rescue service (FRS). The casualty was formally declared deceased by a paramedic from the ambulance service.

All crew of the two boats were breathalysed by the police; all recorded a zero-alcohol result.

A postmortem external examination carried out by a pathologist a few days later confirmed that the casualty had died as a result of a severe head injury.

1.6 BOATS' CREWS

1.6.1 N207

Firefighter Josh Gardener had joined the FRS on 10 September 2018. Prior to this, he had worked at sea on small craft operating in offshore windfarms. He had completed the Royal Yachting Association (RYA) powerboat level 2 training in June 2019.

1.9 PORT AUTHORITY

Milford Haven Port Authority (MHPA) is the statutory harbour authority for Milford Haven waterway and the tidal region of the River Cleddau, including the section of the river where the accident occurred. MHPA is responsible for the navigation, safety and traffic regulation of all vessels navigating within this area. This is achieved through the use of byelaws, General Directions, Special Directions and Notices to Mariners, and included the requirement that all commercial and leisure users within the MHPA area adhere to the International Regulations for Preventing Collisions at Sea 1972 (as amended). Rule 5 of these regulations states:

Every vessel shall at all times maintain a proper look out by sight and hearing as well as by all available means appropriate to the prevailing circumstances and conditions so as to make a full appraisal of the situation and of the risk of collision.

MHPA had designated the area upriver from the Cleddau bridge to the vicinity of Lawrenny as a water-ski area and it was not subject to speed restrictions. Various river areas, including upriver past Lawrenny, were designated as 'Dead Slow Minimum Wake' zones within the MHPA *Leisure Guide (Annex E)*. These minimum wake zones were routinely patrolled by an MHPA water ranger to ensure compliance.

As the local authority responsible for ensuring standards of operations, MHPA generally adopted the MCA coding and certification requirements for commercial vessels operating in its area. Once the coding for *Rescue 1* had lapsed, an agreement had been put in place between MHPA and MWWFRS to continue to operate *Rescue 1* within the MHPA area without the requirement for it to be coded. This agreement had never been extended to cover the SRT boats such as *N207*, and no agreement was in place with any other port authority within the MWWFRS area to allow them to operate without adhering to local commercial craft coding and certification requirements.

1.10 PREVIOUS/SIMILAR ACCIDENTS

1.10.1 Collision between two RIBs *Osprey* and *Osprey II* (MAIB Report No 10/2017⁴)

Two passenger carrying RIBs, *Osprey* and *Osprey II*, collided in the Firth of Forth. A passenger who was sitting on an inflatable tube of *Osprey II* was crushed between *Osprey*'s bow and *Osprey II*'s helm console, resulting in her sustaining serious injuries.

While proceeding in parallel at a speed of around 6 knots (kts), the skipper of each RIB increased speed and commenced a power turn away from each other with the intention of passing each other in the course of completing a round turn. However, as the RIBs turned towards each other it became apparent to both skippers that the RIBs were in danger of colliding. Although they both acted quickly to reduce the speed of their respective vessels and so lessen the impact, they were unable to prevent the collision. The manoeuvre had previously been carried out successfully on several occasions but it had not been formally risk assessed, and no thought had been given regarding what to do if a collision situation developed.

⁴ <https://www.gov.uk/maib-reports/collision-between-rigid-inflatable-boats-osprey-and-osprey-ii-resulting-in-serious-injuries-to-1-passenger>

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 OVERVIEW

The FRS boats *N207* and *Rescue 1* collided because they were carrying out uncoordinated manoeuvres, at speed, in the same stretch of water and neither helmsman realised the risk until it was too late. Firefighter Josh Gardener died because he was struck on the head by *Rescue 1* before or after being knocked out of *N207* into the water.

This section of the report will analyse the circumstances that led to the collision, the reasons why the boats were being operated in the same stretch of water, and the underlying factors that might have contributed to the accident and resultant loss of life.

2.3 THE COLLISION

Once the two boats had left the vicinity of Mill Bay, they were headed for the same agreed location near Lawrenny, with *Rescue 1* ahead and clear of *N207* until its helmsman decided to execute a large full-circle turn to port.

2.3.1 *Rescue 1*

It was unclear why *Rescue 1*'s helmsman decided to undertake the full-circle turn given its proximity to *N207*, rather than proceeding upstream to the agreed rendezvous location. This manoeuvre resulted in the need for *Rescue 1* to subsequently overtake or pass close to *N207*, which was conducting a series of turns under the supervision of the trainer.

The turn to port took *Rescue 1* downstream of *N207*, so it approached the slower boat from behind. When the helmsman saw *N207* heading downstream toward Mill Bay as he executed the final stages of the turn, he thought that it was going to cross clear ahead of him and did not expect *N207* to suddenly turn into his path. The crewman was not keeping an effective lookout and keeping the helmsman informed of the location of *N207*, so as *Rescue 1* came out of its turn and the helmsman looked up, he was surprised to see that collision with *N207* was imminent. The emotional and behavioural response to being surprised might have led to delayed collision avoidance action and disrupted problem solving. His reaction time to implement his decision - to accelerate and turn sharply to starboard - would have been affected by his competence and familiarity with *Rescue 1* and its controls. The qualification to RYA powerboat level 2 should have been sufficient to drive the boat, however the helmsman's lack of practical experience would have diminished his level of competence. Given the speed *Rescue 1* was travelling, it would have been prudent to have planned a greater passing distance from the slower boat, whose exact intentions were unknown.

2.3.2 N207

The three crew of *N207* were aware that *Rescue 1* had accelerated away from Mill Bay and away from their boat. However, their collective attention then returned to the boat handling training and practice for the two firefighters. The positions of the trainer in the middle and the firefighter forward in the boat meant that they could both see forward, to starboard and astern, but neither had an easy view to port without physically moving position. The helmsman steering the boat was seated on the starboard side tube aft and had a view to port, but was focused on practice manoeuvres. As a result, during *Rescue 1*'s turn to port, it went out of the immediate view of the crew of *N207*. Because none of *N207*'s crew were keeping a proper lookout, the helmsman was completely unaware of *Rescue 1*'s position when he initially turned to port and headed downstream.

After seeing the position and direction of travel of *Rescue 1*, the helmsman made the decision to turn further to port to avoid crossing ahead of the approaching RIB. However, because of the closing speed between the two boats, there was insufficient time for the helmsman to bring his boat further to port out of the way of *Rescue 1*.

2.4 LOOKOUT

The lack of effective lookout and awareness of each boat's position led to the collision situation arising.

On *N207* the crew were focused on their training, part of which should have been to nominate a member of the crew to keep a good lookout. With only two boats on the water in that locality, the lookout should have been able to provide the helmsman with information on *Rescue 1*'s position, course and speed. With that information the helmsman would not have turned into the path of *Rescue 1*.

On board *Rescue 1*, the crewman had not been tasked with keeping a lookout and was not paying attention to the activity of *N207*. Had he done so, the risk of collision might have been identified earlier, brought to the attention of the helmsman, and the collision avoided.

In the RYA publication, 'Start Powerboating', written to accompany the powerboat level 2 course, it states:

When intending to travel at higher speeds, you must keep a really good all-round look-out. Check it is clear before you turn and be aware of other water users...

In close proximity and with the relative speeds involved, there was little time to make decisions and take evasive action, especially given the level of competence and experience of most of the crew involved.

2.5 PLANNING AND CONTROL OF THE TRAINING AND FAMILIARISATION EVOLUTIONS

MWWFRS procedures for any training activity required that a full briefing was undertaken prior to commencement. This briefing would be expected to include reference to any specific procedures related to the activity, any guidance and risk assessments, identification of the person in-charge and a full discussion of the intended activity. Because the decision to undertake the training/familiarisation

was agreed on the evening prior to the accident, there was little time to plan it thoroughly. The firefighters had assembled in the morning and held an informal discussion about the activity for the day, but because there was no formal briefing, the MWWFRS's pre-activity planning requirements were not met.

Additionally, although the trainer was identified as being in charge of the training undertaken on *N207*, neither of the crew of *Rescue 1* were nominated as being in charge of that boat, and no individual had been identified as being in overall charge of the activity on-water that morning. Consequently, the morning's activities were not formally planned, and the role of each boat was not discussed.

With the two firefighters planned to undertake the training course in Menai the following day, the purpose and structure of the training and activities being undertaken by *N207* on the day of the accident were clear, and under the control of a qualified trainer. What was unclear was the purpose and intention of the two crew managers crewing *Rescue 1*. Unlike *N207*, the crew of *Rescue 1* had no plan of activities and there was no agreement to support, or be part of, *N207*'s training. Had formal planning and briefing taken place, steps might have been taken to avoid both boats operating at speed in the same part of the river.

2.6 QUALIFICATIONS, CREWING AND TRAINING

2.6.1 Overview

The firefighters' qualification to RYA powerboat level 2 provided them with training that was sufficient to operate boats similar to *N207* and *Rescue 1* as leisure users. However, since the firefighters were at work, they were obliged to comply with the higher qualification standards set by MWWFRS for operating its boats.

MWWFRS's procedures did not distinguish between the crewing requirements to be met when the boats were on operations, and when staff were undergoing training or accruing boat familiarisation time in controlled conditions. These requirements included both the numbers of crew per boat, and the level of qualifications they held. For Milford Haven fire station, which operated two boats, it was difficult to undertake any afloat training or boat familiarisation activities while complying with the operational qualification and crewing requirements. This difficulty was evidenced by the few times that either boat had been used solely for training or familiarisation in the 12 months prior to the accident at Milford Haven.

2.6.2 Training

The Health and Safety Executive defines competence as being:

the combination of training, skills, experience and knowledge that a person has and their ability to apply them to perform a task safely.

The records of training within the MWWFRS pdrPro software recorded only that firefighters had been training with the boats, but did not record how much time any individual spent in charge or helming.

The firefighters had initially been trained to a suitable standard, but the systems were not in place to maintain or build their levels of competence. There was a lack of opportunity to undertake training, to practise and hone boat skills, gain experience

SECTION 3 - CONCLUSIONS

3.1 SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. The firefighter on board *N207* was fatally injured after being struck on the head by *Rescue 1* when the two boats collided. [2.2]
2. *N207* and *Rescue 1* collided because they were operating at speed in close proximity when *N207* turned into the path of *Rescue 1*. [2.2]
3. It is unclear why *Rescue 1*'s helmsman decided to undertake a circular turn to port in the vicinity of *N207*, but given the speed the RIB was travelling it would have been prudent to have planned a greater passing distance from the slower boat, whose exact intentions were unknown. [2.3.1]
4. The crew of *N207* were focused on their boat handling training, and the helmsman was unaware of *Rescue 1*'s position when he turned sharply into its path. Due to the closing speed of the two vessels, his subsequent actions to avoid collision were not effective. [2.3.2]
5. Neither crew was keeping an effective lookout, and so lacked awareness of the two boats' relative positions and movements. [2.4]
6. Mid and West Wales Fire and Rescue Service's pre-activity planning requirements were not met and its SOPs were not followed, with the consequence that: no individual had responsibility for the overall activity; no-one was nominated to be in charge of *Rescue 1*, which had no clear task; *Rescue 1* had insufficient crew, neither of whom held the qualifications required to be the RIB's helmsman; and, no steps were taken to prevent both boats operating in the same stretch of water in an uncoordinated manner. [2.5, 2.6.3]
7. It was difficult for Milford Haven fire station, which operated two boats, to comply with Mid and West Wales Fire and Rescue Service crewing requirements, which reduced the amount of on-water training and familiarisation conducted in the 12 months prior to the accident. [2.6.1]
8. At the time of the accident it is possible that Milford Haven's firefighters were suffering from a gradual erosion of the levels of practical boat handling competence, and a general reduction in their levels of safety awareness for operating their boats; in particular, *Rescue 1*. [2.6.2]
9. Although the firefighters crewing the boats were not wearing the head protection required by AET 5.17, it is unlikely that head protection would have been sufficient to save the firefighter's life when he was struck by *Rescue 1*. [2.7.1]
10. Mid and West Wales Fire and Rescue Service's procedures for boat operations had not been updated for some time, contained misleading information, and made insufficient reference to *Rescue 1* as a pre-determined operational asset. [2.7.2]

11. While the standards for emergency operation will dictate stringent requirements, the ability to train and conduct familiarisation under the same procedures was difficult, and at local level this led to the divergence from Mid and West Wales Fire and Rescue Service's SOPs. [2.7.2]
12. The investigation has found that some port authorities made assumptions that FRS craft operating in their areas complied with an approved standard. However, it was apparent that Mid and West Wales Fire and Rescue Service had overlooked the requirement to operate all of its boats to an approved standard, or that within a local authority or harbour area locally required standards could apply. [2.8]

3.2 SAFETY ISSUES NOT DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. The recording of training and monitoring of competence was not robust enough to identify when a firefighter was helming a boat, and therefore ensure that sufficient experience was logged. [2.6.2]
2. The chance to recognise that shortfalls in training, experience, qualifications and boat certification by Mid and West Wales Fire and Rescue Service was missed during internal audits and inspections and through external peer review. [2.6.3]
3. Mid and West Wales Fire and Rescue Service standard operating procedures did not contain information relevant to the operation of the Zodiac Milpro 470 or the Delta 6.4m RIB. Furthermore, it did not distinguish between requirements for using the boats when staff were undertaking familiarisation, instruction or were on emergency call-out. [2.7.2]
4. The Fire and Rescue Service was not fully aware of the certification requirements for commercially operated boats operating in categorised and non-categorised waters, and hence *Rescue 1* had been operated at sea without the necessary levels of certification. [2.8]
5. Standards for Fire and Rescue Service boats operating in categorised waters and at sea had been overlooked nationally. [2.8]